# Compound Net

*Handling of Domain Fingerprint (Di) – Method and Tendencies*

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## 1. Background and Purpose

This document focuses on the part of the Compound Net project that concerns how information about domain, subgroup, task type and difficulty is best supplied to the combination network. The reference is Compound_Net_Vetenskaplig_Slutrapport_SV_korrekt.docx (November 2026), in which the basic architecture with locked professors and a minimal combination network was established.

The Achilles heel addressed is the method for handling a fingerprint (here called Di) so that it provides a relevant signal without dominating the real data points. The goal is not a finished solution, but to demonstrate that a method exists for handling the problem.

## 2. Starting Point

In the locked architecture the professors receive only the original input X. The combination network receives X + professor outputs + Di. Di is a hierarchical multi-channel fingerprint of fixed length. The original specification was 37 dimensions (8+16+8+1+4).

During the work it was observed that Di's share of the input to the combination network matters. An overly large fingerprint risks crowding out the 12 real data points.

## 3. Tests Performed

### 3.1 Di Size

A reduced Di of 17 dimensions (4+8+4+1) was compared with the original 37-dimensional version. The smaller variant gave lower relative RMSE and better balance against real data.

### 3.2 Gating Alternatives

Several gating variants were tested: fixed domain gating, learned softmax gating, and no gating. In a controlled comparison under the same conditions, Di concatenation (17 dim) performed better than fixed gating.

### 3.3 Results Overview

| Method | Rel. RMSE | Comment |
|---|---|---|
| Di 37 dim (concatenation) | 32.9% | Original |
| **Di 17 dim (concatenation)** | **30.1–33.1%** | Best balance |
| Fixed domain gating | 65.5% | Worse |
| No gating / only X+outputs | ~84% | Baseline |

## 4. What We Have Arrived At

Di concatenation with a balanced dimension (17) is the method that has so far given the best results. It shows that fingerprint information relevant to the task can be exploited by the combination network and yield a clear improvement compared with using only X and professor outputs.

The absolute level (approximately 33% relative RMSE) is still high, but the tendency is clear: how the fingerprint is handled matters. Few parameters that match the data covered by the study are preferable to larger, more complex solutions at this stage.

PFM (Mathematics) was eliminated during the work because it did not reach a reasonable level on its own domain. The system continued with PFE, PFF and PFK.

## 5. Proposed Continuation

- Document 17-dim Di as a working starting point, without locking it as final.
- Test scaled Di or a simple learned compression (4–6 dim) to further reduce the risk of dominance.
- Improve the weaker professors (especially PFF) so that their outputs become more informative.
- Repeat the comparison against a freestanding baseline with the same degree of freedom in order to quantify the parameter relation.
- Keep the focus on the method for fingerprint handling rather than optimising towards a finished product.

## 6. Conclusion

The work shows that a method exists for handling the fingerprint’s Achilles heel. Di concatenation with a balanced size yields a demonstrable tendency toward improvement. This is sufficient as a basis for continued research. An international standard (IDFS) may be considered later, but is not necessary at this stage.